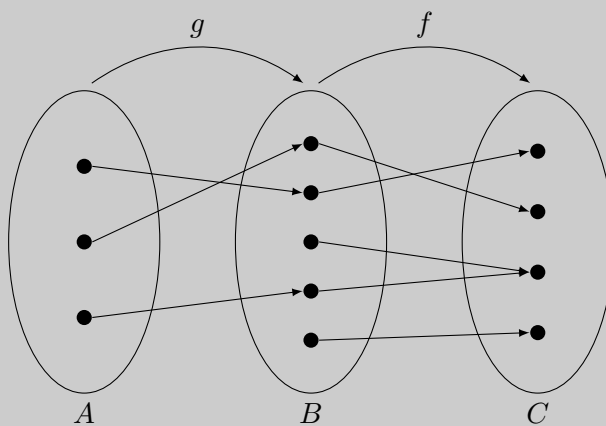


University of Toronto Mississauga
Mathematical and Computational Sciences
MAT102H5S - Term Test 1
Multiple Choice Booklet
Duration - 110 minutes
No Aids Permitted

This booklet contains the multiple choice questions for **Question 5**. Each part of Question 5 includes five multiple choice answers, of which only a single response is correct. **All answers must be indicated on the bubble sheet (page 6) located at the end of the Test Booklet.** *This* booklet will not be collected at the end of the test, and no work in it will be marked.

You will **not** be submitting this booklet at the end of the test, and are therefore free to remove the staple. The pages have been printed single sided, and are intended to be used for any rough work.

The following diagram will be used in MC9



Question 5: Each question is worth 1 mark. Indicate your answer by filling in the bubble sheet on the last page of the exam paper. No part marks will be awarded for this question and you are not required to show your work.

MC 1. Determine which expression should replace ?? to make the following truth table valid.

P	Q	R	??
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	F
F	T	F	F
F	F	T	F
F	F	F	F

- A. $(P \wedge Q) \wedge \neg R$
- B. $P \vee (Q \wedge R)$
- C. $(P \vee Q) \wedge R$
- D. $(P \wedge Q) \vee (P \wedge R)$
- E. None of the above.

MC 2. Define the sets $A = \{n \in \mathbb{Z} : n = k^2 \text{ for some } k \in \mathbb{Z}\}$ and $B = \{n \in \mathbb{Z} : n = k^4 \text{ for some } k \in \mathbb{Z}\}$. Which of the following statements is correct?

- A. $A \subsetneq B$
- B. $B \subsetneq A$
- C. $A = B$
- D. $A \cap B = \emptyset$
- E. None of the above.

MC 3. We say that $g \in \mathbb{Z}$ is *grand* if $\forall a \in \mathbb{Z}, \forall b \in \mathbb{Z}, [g = a \cdot b \implies (a = 1 \vee b = 1)]$. Which of the following defines what it means for g to not be grand?

- A. $\exists a \in \mathbb{Z}, \exists b \in \mathbb{Z}, (g = a \cdot b) \wedge (a \neq 1) \wedge (b \neq 1)$.
- B. $\exists a \in \mathbb{Z}, \exists b \in \mathbb{Z}, [g \neq a \cdot b \implies (a = 1 \vee b = 1)]$.
- C. $\exists a \in \mathbb{Z}, \exists b \in \mathbb{Z}, [g = a \cdot b \implies (a \neq 1 \vee b \neq 1)]$.
- D. $\exists a \in \mathbb{Z}, \exists b \in \mathbb{Z}, [g \neq a \cdot b \vee (a = 1 \vee b = 1)]$.
- E. None of the above.

MC 4. Which of the following statements is/are true?

- (I) $\forall a \in \mathbb{Z}, \forall b \in \mathbb{Z}, a = b^2$ (III) $\exists a \in \mathbb{Z}, \forall b \in \mathbb{Z}, a = b^2$
(II) $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z}, a = b^2$ (IV) $\exists a \in \mathbb{Z}, \exists b \in \mathbb{Z}, a = b^2$.

- A. (II) is the only true statement.
B. (I) and (III) are both true statements.
C. (IV) is the only true statement.
D. (III) and (IV) are both true statements.
E. All of the statements are true.

MC 5. Define a relation $\sim$ on $\mathbb{Z}^+$ by saying that $a \sim b$ if they share at least one digit in common. For example, $123 \sim 205$ as they both share a “2”, while $123 \not\sim 67$. Which of the following statements is correct?

- A. $\sim$ is reflexive, symmetric, and transitive.
B. $\sim$ is reflexive and symmetric, but not transitive.
C. $\sim$ is symmetric and transitive, but not reflexive.
D. $\sim$ is reflexive and transitive, but not symmetric.
E. None of the above.

MC 6. Recall the following weak partial order on $\mathbb{Z} \times \mathbb{Z}$, defined in Assignment 1: We say that $(a, b) \preceq (c, d)$ if $a \leq c$ and $b \leq d$. Define $S = \{(m, n) \in \mathbb{Z} \times \mathbb{Z} : n \geq 0 \text{ and } -2 \leq m \leq -1\}$ and $T = \{(m, n) \in \mathbb{Z} \times \mathbb{Z} : n \geq 0 \text{ and } 1 \leq m \leq 2\}$. What is/are the minimal element(s) of $S \cup T$?

- A. Both $(-2, 0)$ and $(1, 0)$ are minimal .
B. $(-2, 0)$ is the only minimal point.
C. $(1, 0)$ is the only minimal point.
D. $(0, 0)$ is the only minimal point.
E. There are no minimal points.

- MC 7. Define an order relation on $\mathbb{Z}$ by saying that $a \preceq b$ if $a \mid b$. If $S = \{8, 12, 36\}$, what is $\inf(S)$? (Recall that $a \mid b$ if there is an $n \in \mathbb{Z}$ such that $an = b$.)
- A. $\inf(S) = 1$
 - B. $\inf(S) = 8$
 - C. $\inf(S) = 72$
 - D. $\inf(S)$ does not exist.
 - E. None of the above.
- MC 8. Define a relation $\sim$ on $\mathbb{Z}^* \times \mathbb{Z}^*$ by saying that $(a, b) \sim (m, n)$ if $a + m = b + n$. Giovanni wants to prove that $\sim$ is transitive. His proof is given below. On what line, if any, is the first error in his proof?
- A. We want to show that if $(a, b) \sim (b, c)$ and $(b, c) \sim (a, c)$ then $(a, b) \sim (a, c)$.
 - B. By definition, we know that since $(a, b) \sim (b, c)$ then $a + b = b + c$.
 - C. Similarly, since $(b, c) \sim (a, c)$ then $b + c = a + c$.
 - D. Combining the previous two facts, we get that $a + b = b + c = a + c$, which in turn implies $(a, b) \sim (a, c)$, as required. $\square$
 - E. There is no error. The proof is correct.
- MC 9. Suppose that A , B , and C are the sets shown on the front page of this booklet, with $f : B \rightarrow C$ and $g : A \rightarrow B$ as illustrated. Which of the following statements is true of the composite function $f \circ g$?
- A. $f \circ g$ is bijective.
 - B. $f \circ g$ is neither injective nor surjective.
 - C. $f \circ g$ is surjective but not injective.
 - D. $f \circ g$ is injective but not surjective.
 - E. $f \circ g$ is not a function.
- MC 10. Suppose that $f : A \rightarrow B$ is a function and $U \subseteq B$. Which of the following statements is correct?
- A. $U = f(f^{-1}(U))$
 - B. $U \subseteq f(f^{-1}(U))$
 - C. $U \supseteq f(f^{-1}(U))$
 - D. $U \cap f(f^{-1}(U)) = \emptyset$.
 - E. None of the above.